

Amjad Althabteh

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SUMMARY

Performance-focused C++ systems engineer specializing in low-latency architecture, simulation systems, and graphics/rendering pipelines. Focused on GPU throughput, memory layout, cache efficiency, and profiling-driven optimization, from lock-free concurrent systems to custom allocators and rendering engines.

EDUCATION

Wayne State University
B.S. in Computer Science

Detroit, MI | 2025
GPA: 3.76

EXPERIENCE

Immersalab Graphics | Detroit, MI
Graphics Engineer

05/25 – 02/26

- Reduced GPU idle time by ~30% using RenderDoc profiling and rendering pipeline optimizations.
- Designed internal C++/JSON tooling and asset pipeline; profiled and optimized GLSL shader compilation paths.
- Improved frame submission throughput and GPU resource utilization across rendering workloads.

UE Game Programmer C++ | Yorktown, VA
Grand Forge Games

04/26 – 08/26

- Built a gameplay ability system in C++ using Unreal Engine GAS, supporting ability types with Blueprint integration.
- Reduced animation transition overhead using multi-layer blending and reusable C++ systems.

Rare Munchies | Ann Arbor, MI
Software Development Intern

09/23 – 04/24

- Reduced unnecessary re-renders by ~35% through improved state management and component optimization.
- Built reusable TypeScript component systems focused on runtime efficiency and minimal overhead.

Wayne State University | Detroit, MI
Undergraduate Researcher — Orbital Collision Modeling

02/25 – 12/25

- Optimized collision-detection performance by ~30% through spatial partitioning, allowing simulations to scale efficiently across large orbital datasets.
- Designed a spatial-partitioning scheme that reduced simulation runtime by ~30%, improving scalability for large-scale N-body orbital collision analysis.

PROJECTS

Glint 3D Rendering Engine | C++17/OpenGL

Glint-Rendering

- Designed a memory pool allocator using placement new and free list reuse, reducing heap allocation overhead by ~60%.
- Improved rendering performance using frustum culling and GPU instanced rendering, cutting draw calls by ~40%.
- Used L1/L2 cache warming to preload key asset data before rendering and reduce cold-start latency.

IronPath Robotic Control Systems | IronPath Control Systems | Rust

RobicIronPath

- Achieved a 100Hz control loop with ~142μs average latency (p99: 540μs), tracked via custom CSV telemetry and latency..
- Implemented custom allocation tracking using overloaded new/delete operators for memory leak detection.
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3D Space Simulation Engine | C++ | [prediction-model-orbital-collision c++](#)

OrbitalCollision

- Engineered a modular C++17/OpenGL simulation engine simulating 50+ orbital bodies across 1M+ trajectories.
- Reduced simulation runtime by ~40% using spatial partitioning and numerical integration.
- Used loop unrolling and SIMD-friendly vector operations to reduce branching in tight performance loops.

HFT OrderBook | C++20 | Performance System

HFT

- Built a C++20 order matching engine with price-time priority achieving microsecond-level order resolution.
- Eliminated false sharing using cache-aligned (alignas(64)) structures, reducing memory contention under concurrent load.

SKILLS

Languages: C++ (C++11–C++23), Rust, C, C#, Python, TypeScript, Java, Javascript, SQL

Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot, vulkan

Systems: Linux, custom allocators, memory pools, multithreading, lock-free data structures, cache optimization, profiling